

drfpostman

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Create Project

```
django-admin startproject drf_mcq_demo
cd drf_mcq_demo
python manage.py startapp accounts
```

settings.py

```
INSTALLED_APPS = [
    ...
    'rest_framework',      # MCQ 519
    'accounts',
]

REST_FRAMEWORK = {
    "DEFAULT_AUTHENTICATION_CLASSES": [
        "rest_framework_simplejwt.authentication.JWTAuthentication",  # MCQ 506, 518
    ],
    "DEFAULT_PERMISSION_CLASSES": [
        "rest_framework.permissions.AllowAny",  # MCQ 492
    ]
}
```

models.py

```
from django.db import models

class Student(models.Model):
    name = models.CharField(max_length=100)
    email = models.EmailField(unique=True)  # MCQ 513
    age = models.IntegerField()
    city = models.CharField(max_length=100)

    def __str__(self):
        return self.name
```

Database

```
python manage.py makemigrations
python manage.py migrate
python manage.py createsuperuser
```

serializers.py

```
from rest_framework import serializers
from .models import Student
```

```
class StudentSerializer(serializers.ModelSerializer):           # MCQ 476, 477
    class Meta:                                              # MCQ 480
        model = Student
        fields = "__all__"                                   # MCQ 520
        read_only_fields = ['id']                          # MCQ 500

    def validate(self, data):                               # MCQ 502
        if data['age'] < 18:
            raise serializers.ValidationError("Age must be at least 18.")
        return data
```

permissions.py

```
from rest_framework.permissions import BasePermission
```

```
class IsTeacher(BasePermission):                             # MCQ 491, 514
    def has_permission(self, request, view):
        return request.user.is_staff
```

views.py

```
from rest_framework import viewsets
from rest_framework.response import Response                # MCQ 515
from rest_framework.decorators import action                # MCQ 501
from rest_framework.permissions import IsAuthenticated     # MCQ 492
from .models import Student
from .serializers import StudentSerializer

class StudentViewSet(viewsets.ModelViewSet):               # MCQ 481, 482, 522
    queryset = Student.objects.all()
    serializer_class = StudentSerializer
    permission_classes = [IsAuthenticated]                  # MCQ 492, 505, 528

    @action(detail=False, methods=['get'])                  # MCQ 501
```

```

def total(self, request):
    count = Student.objects.count()
    return Response({"Total Students": count}) # MCQ 515, 529

# Function Based API

from rest_framework.decorators import api_view, permission_classes
from rest_framework.permissions import AllowAny
from rest_framework.response import Response

@api_view(['GET']) # MCQ 485, 524
@permission_classes([AllowAny]) # MCQ 492
def welcome(request):
    return Response({"message": "Welcome to DRF"}) # MCQ 515, 529

# Project urls.py

from django.contrib import admin
from django.urls import path, include
from rest_framework.routers import DefaultRouter # MCQ 483
from accounts.views import StudentViewSet, welcome

router = DefaultRouter() # MCQ 483
router.register("students", StudentViewSet) # MCQ 484, 509, 523

urlpatterns = [
    path("admin/", admin.site.urls),
    path("", include(router.urls)), # MCQ 509
    path("welcome/", welcome), # MCQ 485, 524
]

# JWT URLs

from rest_framework_simplejwt.views import (
    TokenObtainPairView,
    TokenRefreshView
)

urlpatterns += [
    path("token/", TokenObtainPairView.as_view()), # MCQ 486, 487, 488
    path("token/refresh/", TokenRefreshView.as_view()), # MCQ 504
]

# serializer.is_valid()

```

```

serializer = StudentSerializer(data=request.data)

if serializer.is_valid():
    serializer.save()
    print(serializer.validated_data)

# Authentication Check

if request.user.is_authenticated:
    return Response({"message": "Logged In"})

# Authorization Header

Authorization: Bearer <access_token>

# HTTP Methods

GET      /students/
POST     /students/
PUT      /students/1/
DELETE   /students/1/

# HTTP Status Codes

200 OK
201 Created
400 Bad Request
401 Unauthorized
403 Forbidden
404 Not Found

# API Versioning

path("api/v1/", include(router.urls))

# REST Best Practices

Use plural URLs (/students/)
Use proper HTTP methods
Return proper status codes

```

MCQ 478, 521

MCQ 479

MCQ 516

MCQ 527

MCQ 489, 490, 526

MCQ 494

MCQ 494

MCQ 495

MCQ 508

MCQ 517

MCQ 496

MCQ 510

MCQ 497

MCQ 493

MCQ 498

MCQ 511

MCQ 499

MCQ 517